Mathematics Stage 3 -- Year B -- Unit 32 2D Space & Area

KLA: Mathematics | Stage 3 | Year 6 | Term: ___________ | Duration: 2-3 weeks | Teacher: _______________ | Year: _______

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency through exploring, applying and communicating mathematical reasoning.
MA3-2DS-01	Investigates and classifies two-dimensional shapes, including triangles and quadrilaterals, based on their properties.
MA3-2DS-02	Selects and uses the appropriate unit to calculate areas, including areas of rectangles.
MA3-2DS-03	Combines, splits and rearranges shapes to determine the area of parallelograms and triangles.
MA4-ARE-C-01	Stage 4 (group A): applies knowledge of area and composite area to solve problems, using the formulas for rectangles, parallelograms and triangles.

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- On-level	Green -- Extension
2D shapes A & transformation terms (recall & vocabulary) Investigate the line and rotational symmetry of quadrilaterals. Identify regular and irregular polygons. Use translate, reflect and rotate to describe transformations; recognise they preserve size. Determine the factors of a whole number.	**Transformations B & Area B (applying & spatial reasoning)** Dissect and rearrange one shape to make another. Find the area of a composite (L-shape) figure in different ways. Transform a parallelogram into a rectangle to find its area; record a method. Compare a triangle's area to a parallelogram with the same base and height. Establish that a triangle is half a parallelogram and record a method for any triangle.	**Extended -- Stage 4 Area formulas** Apply A = lb (rectangle) and A = bh (parallelogram, base x perpendicular height) (MA4-ARE-C-01). Develop and apply A = 1/2 bh for the area of a triangle (MA4-ARE-C-01). Calculate the area of composite figures by dissecting into rectangles, parallelograms and triangles (MA4-ARE-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students describe shapes using symmetry and the language of translate, reflect and rotate, and record methods for finding areas in words.
Understanding and fluency	Students identify symmetry and polygons, dissect and rearrange shapes, and find the areas of composite figures, parallelograms and triangles.
Reasoning	Students reason that transformations preserve size, and that a triangle is half a parallelogram with the same base and height.
Problem solving	Students find areas in more than one way, and (Stage 4) apply area formulas to rectangles, parallelograms, triangles and composite figures.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band placements / key findings	

Post-test / unit evaluation

Date	
Key findings	
Students who need further consolidation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 32 | NESA Mathematics K-10 Syllabus (2022)

Unit 32 -- Outcome Teaching Record

Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Code	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Investigate the line and rotational symmetry of quadrilaterals.	2DS-A 2D shapes: investigate the line and rotational symmetry properties of quadrilaterals MA3-2DS-01 Year A	Suggested: pair with R2 **Mini Lesson 1**		
R2	Identify regular and irregular polygons.	2DS-A 2D shapes: identify regular and irregular polygons MA3-2DS-01 Year A	Suggested: pair with R1 **Mini Lesson 2**		
R3	Describe translations, reflections and rotations; recognise they preserve size.	2DS-B 2D shapes: use translate/reflect/rotate to describe transformations; transformations change position/orientation but not size MA3-2DS-01 Year B	**Mini Lesson 3**		
R4	Determine the factors of a whole number.	MR-A: determine factors for a given whole number (supports dissecting composite areas) MA3-MR-01 Year A	**Mini Lesson 5**		
Y1	Dissect and rearrange one shape to make another.	2DS-B 2D shapes: dissect and rearrange one shape to make another MA3-2DS-01 Year B	**Mini Lesson 4**		
Y2	Find the area of a composite (L-shape) figure in different ways.	2DS-B Area: find different ways to calculate the area of a composite L-shape figure MA3-2DS-02 / MA3-2DS-03 Year B	**Mini Lesson 6**		
Y3	Transform a parallelogram into a rectangle to find its area; record a method.	2DS-B Area: show how to transform a parallelogram into a rectangle to find its area; record a method for any parallelogram MA3-2DS-03 Year B	Suggested: pair with Y4/Y5 **Mini Lesson 7**		
Y4	Compare a triangle's area to a parallelogram with the same base and height.	2DS-B Area: investigate the area of a triangle by comparing it to a parallelogram with the same base length and height MA3-2DS-03 Year B	Suggested: pair with Y5 **Mini Lesson 8**		
Y5	Establish that a triangle is half a parallelogram; record a method for any triangle.	2DS-B Area: establish the triangle-parallelogram relationship (duplicate and rotate); record a method for any triangle MA3-2DS-03 Year B	Suggested: pair with Y4 **Mini Lesson 9**		
G1	Apply the area formulas for a rectangle (A = lb) and a parallelogram (A = bh).	Stage 4 ARE-A: A = lb (rectangle/square); A = bh (parallelogram, base x perpendicular height) MA4-ARE-C-01 Stage 4 (group A)	**Mini Lesson 10**		
G2	Develop and apply the area formula for a triangle (A = 1/2 bh).	Stage 4 ARE-A: develop and apply A = 1/2 bh for the area of a triangle MA4-ARE-C-01 Stage 4 (group A)	**Mini Lesson 11**		
G3	Calculate the area of composite figures by dissecting into rectangles, parallelograms and triangles.	Stage 4 ARE-A: area of composite figures dissected into rectangles/squares/parallelograms/triangles MA4-ARE-C-01 Stage 4 (group A)	**Mini Lesson 12**		

 2DS = Two-dimensional spatial structure (shapes, transformations, area). MR = Multiplicative Relations (factors). ARE = Area (Stage 4). A/B = syllabus content groups (A classify shapes & symmetry; B transform & find area by rearrangement, taught as home content).

Unit 32 -- Mini Lesson Sequence

2D Space & Area | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 1-5	Yellow band -- On-level -- Mini Lessons 4-9	Green band -- Extension -- Mini Lessons 10-12

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 32. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Symmetry of quadrilaterals **[HANDS-ON]**

R1 | Red band | DoE Lesson 1 (adapted)

Duration: 25-30 min	**Outcomes: R1**	Syllabus: MA3-2DS-01, MAO-WM-01 | 2DS-A: line and rotational symmetry of quadrilaterals

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Line and rotational symmetry | Key resources: Quadrilateral cut-outs, mirrors, tracing paper, paper to fold

Physical materials needed: quadrilateral cut-outs, mirrors, tracing paper

Learning intention	Success criteria
Students are learning to: find the lines of symmetry and the order of rotational symmetry of quadrilaterals	Students can: * I can find the lines of symmetry of a square, rectangle and other quadrilaterals by folding. * I can turn a shape to find its order of rotational symmetry. * I can compare the symmetry of different quadrilaterals.

Mini masterclass structure (25-30 min)

Activate (4 min): Fold a square to find its lines of symmetry. Ask how many it has, and whether a parallelogram has any.

Model (10 min): Find lines of symmetry by folding, then rotational symmetry by tracing a shape and turning it to count how many times it matches in a full turn.

Guided practice (10 min): Students investigate the line and rotational symmetry of a set of quadrilaterals, recording lines of symmetry and order of rotational symmetry.

Check (3 min): Exit question: how many lines of symmetry does a rectangle have, and what is its order of rotational symmetry?

Key vocabulary	Differentiation
symmetry, line of symmetry, rotational symmetry, order of rotational symmetry, quadrilateral, fold, turn	**Support:** Use a mirror and pre-cut shapes; fold to find lines of symmetry before rotating. **Extension:** Compare the symmetry of all the special quadrilaterals and order them by number of lines of symmetry.

Assessment	Teaching notes (annotate here)

Students investigate line and rotational symmetry of quadrilaterals — links to R1 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Regular and irregular polygons

R2 | Red band | DoE Lesson 1 (adapted)

Duration: 20-25 min	**Outcomes: R2**	Syllabus: MA3-2DS-01, MAO-WM-01 | 2DS-A: identify regular and irregular polygons

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Line and rotational symmetry (regular & irregular polygons) | Key resources: Polygon cards, rulers, protractors

Learning intention	Success criteria
Students are learning to: identify polygons as regular or irregular	Students can: * I know a regular polygon has all sides and all angles equal. * I can identify a polygon as regular or irregular. * I can name polygons by their number of sides.

Mini masterclass structure (20-25 min)

Activate (3 min): Show a square and a non-square rectangle. Ask which is 'regular' and why.

Model (9 min): Define regular (all sides and angles equal). Sort polygons into regular and irregular, checking sides and angles.

Guided practice (8 min): Students sort a set of polygons into regular and irregular and justify each choice.

Check (3 min): Exit question: is a rhombus a regular polygon? Explain.

Key vocabulary	Differentiation
polygon, regular, irregular, sides, angles, equal, pentagon, hexagon, octagon	**Support:** Provide a checklist (all sides equal? all angles equal?) to test each polygon. **Extension:** Explain why a rhombus and a rectangle are irregular even though they look symmetric.

Assessment	Teaching notes (annotate here)
Students identify regular and irregular polygons with justification — links to R2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 3 Translate, reflect and rotate [HANDS-ON]

R3 | Red band | DoE Lesson 2 (adapted)

Duration: 25-30 min	**Outcomes: R3**	Syllabus: MA3-2DS-01, MAO-WM-01 | 2DS-B: terms translate/reflect/rotate; transformations preserve size

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Transformation changes position and orientation but not size | Key resources: Cut-out shapes, grid paper, mirror

Physical materials needed: cut-out shapes, grid paper, mirror

Learning intention	Success criteria
Students are learning to: use the terms translate, reflect and rotate to describe transformations and recognise they do not change size	Students can: * I can use translate (slide), reflect (flip) and rotate (turn) to describe how a shape has moved. * I know a transformation changes a shape's position and orientation. * I know a translation, reflection or rotation does not change a shape's size.

Mini masterclass structure (25-30 min)

Activate (4 min): Slide, flip and turn a cut-out shape. Ask students to name each move.

Model (10 min): Model translate, reflect and rotate on grid paper, naming each. Check the shape is the same size before and after each transformation.

Guided practice (10 min): Students perform and label translations, reflections and rotations of a shape and confirm the size is unchanged.

Check (3 min): Exit question: name the transformation that flips a shape over a line, and state what stays the same.

Key vocabulary	Differentiation
translate, slide, reflect, flip, rotate, turn, transformation, position, orientation, size	**Support:** Use physical cut-out shapes to perform each move before describing it. **Extension:** Describe a sequence of two transformations that returns a shape to its start.

Assessment	Teaching notes (annotate here)
Students describe transformations and recognise they preserve size — links to R3 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 4 Dissecting and rearranging shapes [HANDS-ON]

Y1 | Yellow band | DoE Lesson 3 (adapted)

Duration: 30 min	**Outcomes: Y1**	Syllabus: MA3-2DS-01, MAO-WM-01 | 2DS-B: dissect and rearrange one shape to make another

Linked to: DoE Lesson 3 (adapted) -- Lesson 3 -- Dissect and rearrange one shape to make another | Key resources: Shapes to cut, scissors, grid paper, tangram sets

Physical materials needed: shapes to cut, scissors, tangram sets, grid paper

Learning intention	Success criteria
Students are learning to: dissect a shape and rearrange the pieces using transformations to make another shape	Students can: * I can cut a shape into pieces. * I can translate, reflect and rotate the pieces to make a new shape. * I know the total area stays the same when I rearrange the pieces.

Mini masterclass structure (30 min)

Activate (4 min): Cut a rectangle along a diagonal. Ask what shapes the two pieces could make when rearranged.

Model (10 min): Dissect a shape and rearrange the pieces (sliding, flipping, turning) to form another shape, noting the area is conserved.

Guided practice (12 min): Students dissect shapes and rearrange them to make a target shape (tangram-style), describing the transformations used.

Check (4 min): Exit question: cut a square into two pieces and rearrange them into a triangle — what stays the same?

Key vocabulary	Differentiation
dissect, rearrange, transformation, translate, reflect, rotate, area, conserved	**Support:** Provide pre-cut pieces and a target outline to fill. **Extension:** Find more than one way to dissect and rearrange a shape into the same target.

Assessment	Teaching notes (annotate here)
Students dissect and rearrange shapes using transformations — links to Y1 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 5 Factors of a whole number

R4 | Red band | DoE Lesson 5 (adapted)

Duration: 20-25 min	**Outcomes: R4**	Syllabus: MA3-MR-01, MAO-WM-01 | MR-A: determine factors for a given whole number

Linked to: DoE Lesson 5 (adapted) -- Lesson 5 -- Area of composite shapes (using factors) | Key resources: Counters or square tiles, grid paper

Learning intention	Success criteria
Students are learning to: find all the factors of a given whole number	Students can: * I can find factor pairs of a number. * I can list all the factors of a whole number. * I can use factors to find possible side lengths of a rectangle with a given area.

Mini masterclass structure (20-25 min)

Activate (3 min): Arrange 12 tiles into rectangles. Ask what side lengths are possible (factor pairs of 12).

Model (9 min): List the factors of 24 by finding factor pairs (1x24, 2x12, 3x8, 4x6). Connect to possible rectangle dimensions for area 24.

Guided practice (10 min): Students list all factors of several numbers and link them to rectangle dimensions for a given area.

Check (3 min): Exit question: list all the factors of 18.

Key vocabulary	Differentiation
factor, factor pair, multiple, whole number, dimensions, area	**Support:** Use tiles to build rectangles and read off factor pairs. **Extension:** Find numbers with exactly two factors (primes) and numbers with many factors.

Assessment	Teaching notes (annotate here)
Students determine all factors of a whole number — links to R4 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 6 Area of composite L-shapes [HANDS-ON]

Y2 | Yellow band | DoE Lessons 4-5 (adapted)

Duration: 30 min	**Outcomes: Y2**	Syllabus: MA3-2DS-02, MA3-2DS-03, MAO-WM-01 | 2DS-B Area: find different ways to calculate the area of a composite L-shape

Linked to: DoE Lessons 4-5 (adapted) -- Lessons 4-5 -- Area of a composite L-shape figure | Key resources: Grid paper, L-shape figures, rulers

Physical materials needed: grid paper, L-shape figures, rulers

Learning intention	Success criteria
Students are learning to: find the area of a composite L-shape figure in more than one way	Students can: * I can split an L-shape into two rectangles and add their areas. * I can surround an L-shape to make a big rectangle and subtract the missing piece. * I can show the area is the same using both methods.

Mini masterclass structure (30 min)

Activate (4 min): Show an L-shaped figure on a grid. Ask how we could split it to find its area.

Model (10 min): Find the area two ways: (1) split into two rectangles and add; (2) complete the big rectangle and subtract the cut-out. Confirm both give the same area.

Guided practice (12 min): Students find the area of several composite L-shapes using both the split and surround-and-subtract methods.

Check (4 min): Exit question: find the area of this L-shape two different ways.

Key vocabulary	Differentiation
composite figure, L-shape, area, rectangle, split, surround, subtract, square units	**Support:** Provide grid paper to count squares; pre-mark the split line. **Extension:** Find the area of composite shapes made from three or more rectangles.

Assessment	Teaching notes (annotate here)
Students find composite-figure areas in more than one way — links to Y2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 7 Area of a parallelogram [HANDS-ON]

Y3 | Yellow band | DoE Lesson 6 (adapted)

Duration: 30 min	**Outcomes: Y3**	Syllabus: MA3-2DS-03, MAO-WM-01 | 2DS-B Area: transform a parallelogram into a rectangle; record a method

Linked to: DoE Lesson 6 (adapted) -- Lesson 6 -- Transform a parallelogram to find its area | Key resources: Parallelogram cut-outs, scissors, rulers, grid paper

Physical materials needed: parallelogram cut-outs, scissors, rulers, grid paper

Learning intention	Success criteria
Students are learning to: transform a parallelogram into a rectangle to find its area and record a method	Students can: * I can cut a right-angled triangle off one end of a parallelogram and move it to the other end to make a rectangle. * I can find the area of the rectangle (and so the parallelogram). * I can record a method: area = base x perpendicular height.

Mini masterclass structure (30 min)

Activate (4 min): Show a parallelogram on grid paper. Ask how we could turn it into a rectangle to find its area.

Model (10 min): Cut a right triangle off one end, translate it to the other end to form a rectangle, and find the area. Record 'area = base x perpendicular height'.

Guided practice (12 min): Students transform parallelograms into rectangles to find their areas and write the method in words.

Check (4 min): Exit question: a parallelogram has base 6 cm and perpendicular height 4 cm — what is its area?

Key vocabulary	Differentiation
parallelogram, rectangle, base, perpendicular height, area, transform, rearrange	**Support:** Provide pre-cut parallelograms with the cut line marked. **Extension:** Explain why the slant side is not used in the area, only the perpendicular height.

Assessment	Teaching notes (annotate here)
Students find the area of a parallelogram by rearrangement and record a method — links to Y3 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Comparing a triangle to a parallelogram [HANDS-ON]

Y4 | Yellow band | DoE Lesson 7 (adapted)

Duration: 25-30 min	**Outcomes: Y4**	Syllabus: MA3-2DS-03, MAO-WM-01 | 2DS-B Area: compare a triangle's area to a parallelogram with the same base and height

Linked to: DoE Lesson 7 (adapted) -- Lesson 7 -- Determine the area of a triangle | Key resources: Matching triangle and parallelogram cut-outs, grid paper

Physical materials needed: matching triangle & parallelogram cut-outs, grid paper

Learning intention	Success criteria
Students are learning to: investigate the area of a triangle by comparing it to a parallelogram with the same base and height	Students can: * I can place a triangle on a parallelogram with the same base and height. * I can see the triangle covers half the parallelogram. * I can explain the triangle's area is half the parallelogram's.

Mini masterclass structure (25-30 min)

Activate (4 min): Show a triangle and a parallelogram with the same base and height. Ask how their areas compare.

Model (10 min): Overlay the triangle on the parallelogram (same base and height). Show two copies of the triangle exactly cover the parallelogram, so the triangle is half.

Guided practice (10 min): Students compare several triangles with same-base/height parallelograms and record that the triangle is half the area.

Check (3 min): Exit question: a parallelogram has area 20 cm2 — what is the area of a triangle with the same base and height?

Key vocabulary	Differentiation
triangle, parallelogram, base, height, half, area, compare	**Support:** Use matching cut-outs students can physically overlay. **Extension:** Show the half relationship holds for an obtuse triangle too.

Assessment	Teaching notes (annotate here)
Students relate a triangle's area to a same-base/height parallelogram — links to Y4 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 9 Area of a triangle: establishing the method [HANDS-ON]

Y5 | Yellow band | DoE Lesson 8 (adapted)

Duration: 30 min	**Outcomes: Y5**	Syllabus: MA3-2DS-03, MAO-WM-01 | 2DS-B Area: triangle-parallelogram relationship (duplicate & rotate); record a method

Linked to: DoE Lesson 8 (adapted) -- Lesson 8 -- Relationship between the area of a triangle and a parallelogram | Key resources: Triangle cut-outs (two copies each), scissors, grid paper

Physical materials needed: triangle cut-outs (two copies each), scissors, grid paper

Learning intention	Success criteria
Students are learning to: establish that a triangle is half a parallelogram and record a method for finding the area of any triangle	Students can: * I can duplicate and rotate a triangle to form a parallelogram. * I know the triangle is half that parallelogram. * I can record a method: area of a triangle = half x base x perpendicular height.

Mini masterclass structure (30 min)

Activate (4 min): Take two identical triangles. Ask how to fit them together to make a parallelogram.

Model (10 min): Duplicate and rotate a triangle to form a parallelogram (base x height). The triangle is half, so area = 1/2 x base x perpendicular height. Record in words.

Guided practice (12 min): Students build parallelograms from pairs of triangles, then calculate triangle areas using the half-base-height method.

Check (4 min): Exit question: a triangle has base 8 cm and perpendicular height 5 cm — what is its area?

Key vocabulary	Differentiation
triangle, parallelogram, duplicate, rotate, base, perpendicular height, half, area, method	**Support:** Provide two pre-cut identical triangles to join into a parallelogram. **Extension:** Record the method as a formula and use it for triangles in different orientations.

Assessment	Teaching notes (annotate here)
Students establish and record the area-of-a-triangle method — links to Y5 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 10 Area formulas for rectangles and parallelograms (Stage 4)

G1 | Green band | Original mini lesson

Duration: 30-35 min	**Outcomes: G1**	Syllabus: MA4-ARE-C-01, MAO-WM-01 | Stage 4 ARE-A: A = lb (rectangle); A = bh (parallelogram)

Learning intention	Success criteria
Students are learning to: apply the area formulas for a rectangle and a parallelogram	Students can: * I can apply A = l x b to find the area of a rectangle. * I can apply A = b x h to find the area of a parallelogram. * I can solve area problems using these formulas.

Mini masterclass structure (30-35 min)

Activate (4 min): Recall area = base x perpendicular height for a parallelogram (Mini Lesson 7). Write it as a formula.

Model (11 min): Apply A = lb to a rectangle (8 cm x 5 cm = 40 cm2) and A = bh to a parallelogram (base 7 cm, height 4 cm = 28 cm2). Stress 'perpendicular height', not the slant side.

Guided practice (12 min): Students apply the formulas to a set of rectangles and parallelograms and solve a word problem.

Check (5 min): Exit question: find the area of a parallelogram with base 9 cm and perpendicular height 6 cm.

Key vocabulary	Differentiation
area, formula, rectangle, parallelogram, length, breadth, base, perpendicular height, square units	**Support:** Provide the formula and a labelled diagram; identify the perpendicular height first. **Extension:** Find a missing base or height given the area; compare parallelograms with the same area.

Assessment	Teaching notes (annotate here)
Students apply the rectangle and parallelogram area formulas — links to G1 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 11 Area formula for a triangle (Stage 4)

G2 | Green band | Original mini lesson

Duration: 30 min	**Outcomes: G2**	Syllabus: MA4-ARE-C-01, MAO-WM-01 | Stage 4 ARE-A: develop and apply A = 1/2 bh

Learning intention	Success criteria
Students are learning to: develop and apply the formula for the area of a triangle	Students can: * I know a triangle is half a parallelogram with the same base and height. * I can apply A = 1/2 x b x h to find the area of a triangle. * I can solve triangle area problems.

Mini masterclass structure (30 min)

Activate (4 min): Recall the triangle is half a parallelogram (Mini Lesson 9). Write the formula A = 1/2 bh.

Model (11 min): Apply A = 1/2 bh to a triangle (base 10 cm, height 6 cm = 30 cm2). Repeat for a right and an obtuse triangle, identifying the perpendicular height each time.

Guided practice (11 min): Students apply A = 1/2 bh to a set of triangles in different orientations and solve a word problem.

Check (4 min): Exit question: find the area of a triangle with base 12 cm and perpendicular height 5 cm.

Key vocabulary	Differentiation
triangle, area, formula, base, perpendicular height, half, square units	**Support:** Provide the formula and a labelled triangle; mark the perpendicular height. **Extension:** Find a missing base or height given the area; find the area of a triangle on a grid by reading base and height.

Assessment	Teaching notes (annotate here)
Students apply A = 1/2 bh to find triangle areas — links to G2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 12 Composite areas with formulas (Stage 4)

G3 | Green band | Original mini lesson

Duration: 30-35 min	**Outcomes: G3**	Syllabus: MA4-ARE-C-01, MAO-WM-01 | Stage 4 ARE-A: area of composite figures by dissection

Learning intention	Success criteria
Students are learning to: calculate the area of composite figures by dissecting them into rectangles, parallelograms and triangles	Students can: * I can dissect a composite figure into rectangles, parallelograms and triangles. * I can find the area of each part using a formula. * I can add the parts to find the total area.

Mini masterclass structure (30-35 min)

Activate (4 min): Show a composite shape made of a rectangle and a triangle. Ask how to split it to use formulas.

Model (11 min): Dissect a composite figure into a rectangle and a triangle, find each area with a formula, and add them. Then a shape needing subtraction.

Guided practice (13 min): Students dissect and find the area of several composite figures using the area formulas, recording each part.

Check (5 min): Exit question: find the area of a composite shape made of a 6x4 rectangle and a triangle (base 6, height 3).

Key vocabulary	Differentiation
composite figure, dissect, rectangle, parallelogram, triangle, area, formula, total	**Support:** Pre-mark the dissection lines; provide the formulas. **Extension:** Solve composite-area word problems involving real contexts (gardens, floor plans).

Assessment	Teaching notes (annotate here)
Students find composite areas by dissecting and applying formulas — links to G3 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Unit 32 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Investigate the line and rotational symmetry of quadrilaterals.		
Video	Intro to Symmetry (Part 1): Lines of Symmetry | Math with Mr. J	https://www.youtube.com/watch?v=dAqDwuHOi4g
Video	Intro to Symmetry (Part 2): Lines of Symmetry | Math with Mr. J	https://www.youtube.com/watch?v=Ae6kn4BqqrA
Video	Line and Rotational Symmetry | Mr Morley Maths	https://www.youtube.com/watch?v=RnqgIR8pljg
Worksheet (print)	Corbettmaths — Symmetry (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/symmetry-pdf.pdf
R2 -- Identify regular and irregular polygons.		
Video	What is a Polygon? An Introductory Guide | Math with Mr. J	https://www.youtube.com/watch?v=mGrst2zkCl8
Video	How Do You Identify Regular and Irregular Polygons? (KS2) | Twinkl	https://www.youtube.com/watch?v=osr-fLanREs
Video	Regular & Irregular Polygons | IconMath	https://www.youtube.com/watch?v=UN57vvrMrY4
R3 -- Describe translations, reflections and rotations; recognise they preserve size.		
Video	3 Types of Transformations: Translations, Reflections & Rotations | Turtlediary	https://www.youtube.com/watch?v=VJTxv-tRKj0
Video	Transformations: Translations, Reflections and Rotations | Math Defined	https://www.youtube.com/watch?v=YD3HIMUae_4
Worksheet (print)	Corbettmaths — Translations (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/translations-pdf.pdf
Worksheet (print)	Corbettmaths — Reflections (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/reflections-pdf.pdf
R4 -- Determine the factors of a whole number.		
Video	Finding Factors of Numbers: What are Factors? | Primary School Made Easy	https://www.youtube.com/watch?v=f6LhwjjOsbg
Video	What are Factors and How are They Used? (KS2) | Twinkl	https://www.youtube.com/watch?v=rJcxEktyuRY
Worksheet (print)	Corbettmaths — Factors (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Factors-216-pdf.pdf
Y1 -- Dissect and rearrange one shape to make another.		
Video	3 Types of Transformations: Translations, Reflections & Rotations | Turtlediary	https://www.youtube.com/watch?v=VJTxv-tRKj0
Video	Transformations: Translations, Reflections and Rotations | Math Defined	https://www.youtube.com/watch?v=YD3HIMUae_4
Worksheet (print)	Corbettmaths — Rotations (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/rotations-pdf.pdf
Y2 -- Find the area of a composite (L-shape) figure in different ways.		
Video	Finding the Area of a Composite Figure (Rectangles) | Math with Mr. J	https://www.youtube.com/watch?v=z4Lat1uOQI4
Video	Finding the Area of Composite Figures with Triangles | Math with Mr. J	https://www.youtube.com/watch?v=bo483Varm-U
Worksheet (print)	Corbettmaths — Area of an L-Shape (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-an-l-shape-pdf.pdf
Worksheet (print)	Corbettmaths — Area of Compound Shapes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-compound-shapes-pdf.pdf
Y3 -- Transform a parallelogram into a rectangle to find its area; record a method.		
Video	Area of a Parallelogram and a Triangle | Virtual Elementary School	https://www.youtube.com/watch?v=l9ZagUlu8T4
Video	Find the Area of Triangles and Parallelograms | your math tutor	https://www.youtube.com/watch?v=IXxBaMWUUR4
Worksheet (print)	Corbettmaths — Area of a Parallelogram (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-parallelogram-pdf.pdf
Y4 -- Compare a triangle's area to a parallelogram with the same base and height.		
Video	Area of a Parallelogram and a Triangle | Virtual Elementary School	https://www.youtube.com/watch?v=l9ZagUlu8T4

Type	Resource	Link
Video	Find the Area of Triangles and Parallelograms | your math tutor	https://www.youtube.com/watch?v=IXxBaMWUUR4
Worksheet (print)	Corbettmaths — Area of a Triangle (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-triangle-pdf.pdf
Y5 -- Establish that a triangle is half a parallelogram; record a method for any triangle.		
Video	How to Find Area: Rectangles, Triangles & More | Math with Mr. J	https://www.youtube.com/watch?v=LpyzdO2fXtA
Video	Area of a Parallelogram and a Triangle | Virtual Elementary School	https://www.youtube.com/watch?v=l9ZagUlu8T4
Worksheet (print)	Corbettmaths — Area of a Triangle (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-triangle-pdf.pdf
G1 -- Apply the area formulas for a rectangle (A = lb) and a parallelogram (A = bh).		
Video	How to Find Area: Rectangles, Triangles & More | Math with Mr. J	https://www.youtube.com/watch?v=LpyzdO2fXtA
Video	Area of a Parallelogram and a Triangle | Virtual Elementary School	https://www.youtube.com/watch?v=l9ZagUlu8T4
Worksheet (print)	Corbettmaths — Area of a Rectangle (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-rectangle-pdf.pdf
Worksheet (print)	Corbettmaths — Area of a Parallelogram (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-parallelogram-pdf.pdf
G2 -- Develop and apply the area formula for a triangle (A = 1/2 bh).		
Video	Area of a Parallelogram and a Triangle | Virtual Elementary School	https://www.youtube.com/watch?v=l9ZagUlu8T4
Video	Find the Area of Triangles and Parallelograms | your math tutor	https://www.youtube.com/watch?v=IXxBaMWUUR4
Worksheet (print)	Corbettmaths — Area of a Triangle (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-triangle-pdf.pdf
G3 -- Calculate the area of composite figures by dissecting into rectangles, parallelograms and triangles.		
Video	Finding the Area of Composite Figures with Triangles | Math with Mr. J	https://www.youtube.com/watch?v=bo483Varm-U
Video	Finding the Area of a Composite Figure (Rectangles) | Math with Mr. J	https://www.youtube.com/watch?v=z4Lat1uOQI4
Worksheet (print)	Corbettmaths — Area of Compound Shapes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-compound-shapes-pdf.pdf
Worksheet (print)	Corbettmaths — Area of an L-Shape (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-an-l-shape-pdf.pdf
Beyond SOLO -- Stage 4 extension		
Video	How to Find Area: Rectangles, Triangles & More | Math with Mr. J	https://www.youtube.com/watch?v=LpyzdO2fXtA
Video	Area of a Parallelogram and a Triangle | Virtual Elementary School	https://www.youtube.com/watch?v=l9ZagUlu8T4
Video	Finding the Area of Composite Figures with Triangles | Math with Mr. J	https://www.youtube.com/watch?v=bo483Varm-U
Worksheet (print)	Corbettmaths — Area of a Parallelogram (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-parallelogram-pdf.pdf
Worksheet (print)	Corbettmaths — Area of a Triangle (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-triangle-pdf.pdf
Worksheet (print)	Corbettmaths — Area of Compound Shapes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-compound-shapes-pdf.pdf
Website	Maths is Fun — Area (formulas & interactive)	https://www.mathsisfun.com/geometry/area.html
Website	Maths is Fun — Parallelogram (interactive)	https://www.mathsisfun.com/geometry/parallelogram.html
Website	Maths is Fun — Area of Irregular Polygons (interactive)	https://www.mathsisfun.com/geometry/area-irregular-polygons.html
Website	Maths is Fun — Symmetry (interactive)	https://www.mathsisfun.com/geometry/symmetry.html

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- [ ] Corbettmaths — Symmetry (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/symmetry-pdf.pdf
- [ ] Corbettmaths — Translations (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/translations-pdf.pdf
- [ ] Corbettmaths — Reflections (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/reflections-pdf.pdf
- [ ] Corbettmaths — Factors (PDF) -- https://corbettmaths.com/wp-content/uploads/2018/11/Factors-216-pdf.pdf

- ☐ Corbettmaths — Rotations (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/rotations-pdf.pdf
- ☐ Corbettmaths — Area of an L-Shape (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/area-of-an-l-shape-pdf.pdf
- ☐ Corbettmaths — Area of Compound Shapes (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/area-of-compound-shapes-pdf.pdf
- ☐ Corbettmaths — Area of a Parallelogram (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-parallelogram-pdf.pdf
- ☐ Corbettmaths — Area of a Triangle (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-triangle-pdf.pdf
- ☐ Corbettmaths — Area of a Rectangle (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-rectangle-pdf.pdf